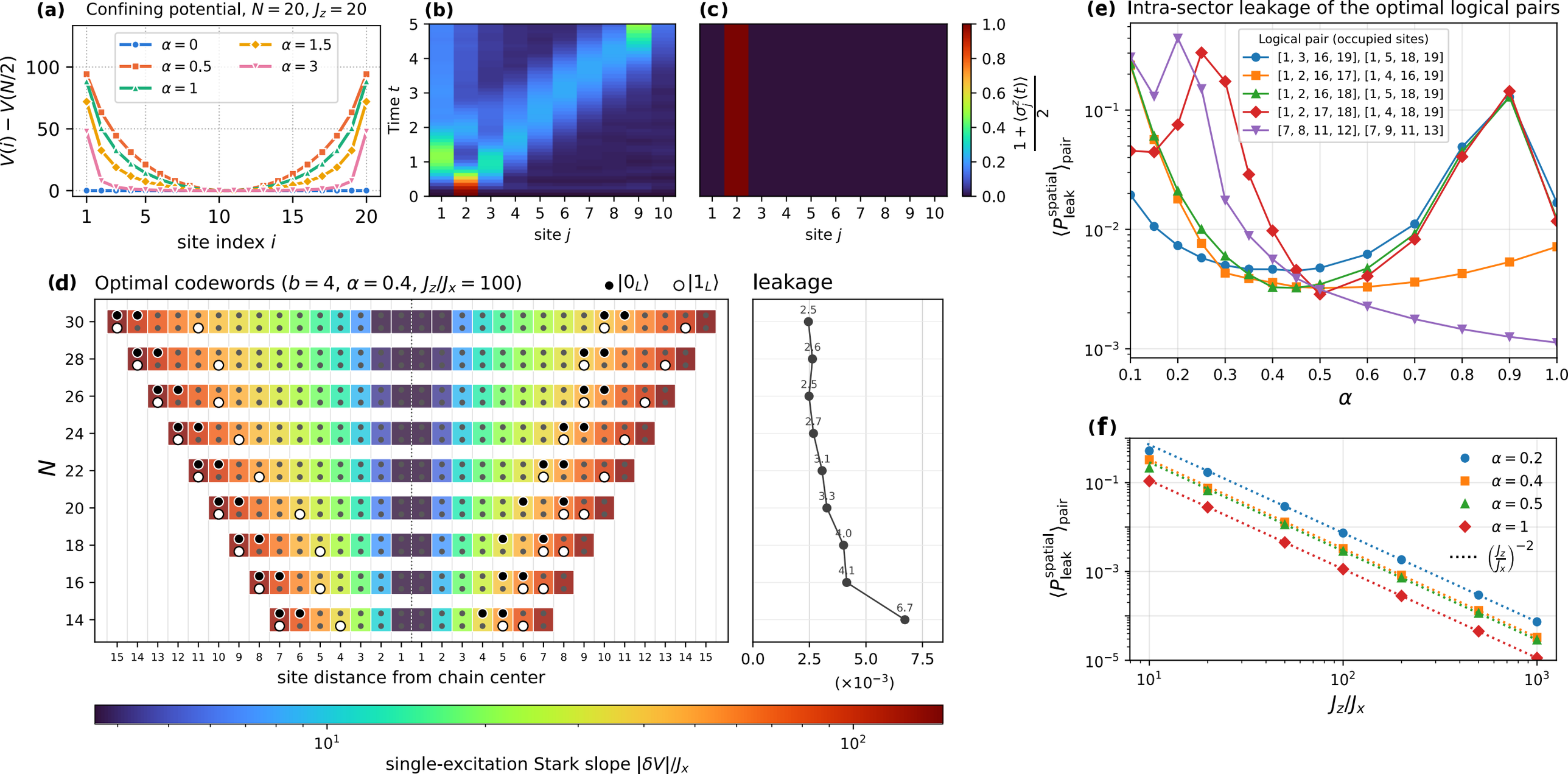

(a)
Confining potential, $N = 20$, $J_z = 20$
$V(i) - V(N/2)$
site index i
$\alpha = 0$
$\alpha = 0.5$
$\alpha = 1$
$\alpha = 1.5$
$\alpha = 3$
(b)
Time t
site j
(c)
site j
$\frac{1 + \langle \sigma_j^z(t) \rangle}{2}$
(d)
Optimal codewords ($b = 4$, $\alpha = 0.4$, $J_z/J_x = 100$)
$\bullet |0_L\rangle$ $\circ |1_L\rangle$
N
site distance from chain center
leakage
$(\times 10^{-3})$
single-excitation Stark slope $|\delta V|/J_x$
(e)
Intra-sector leakage of the optimal logical pairs
Logical pair (occupied sites)
[1, 3, 16, 19], [1, 5, 18, 19]
[1, 2, 16, 17], [1, 4, 16, 19]
[1, 2, 16, 18], [1, 5, 18, 19]
[1, 2, 17, 18], [1, 4, 18, 19]
[7, 8, 11, 12], [7, 9, 11, 13]
$\langle P_{\text{leak}}^{\text{spatial}} \rangle_{\text{pair}}$
α
(f)
$\langle P_{\text{leak}}^{\text{spatial}} \rangle_{\text{pair}}$
$\alpha = 0.2$
$\alpha = 0.4$
$\alpha = 0.5$
$\alpha = 1$
$\left(\frac{J_z}{J_x}\right)^{-2}$
J_z/J_x